Problem 6

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Problem 1 *Two functions, h and g, are given*

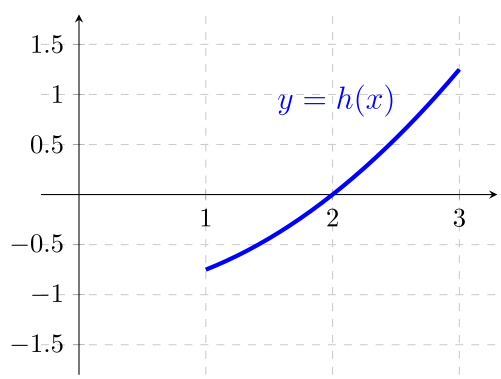
$$h(x) = \frac{x^2 - 4}{4},\ 1 < x < 3$$

$$g(x) = x - 2,\ 1 < x < 3$$

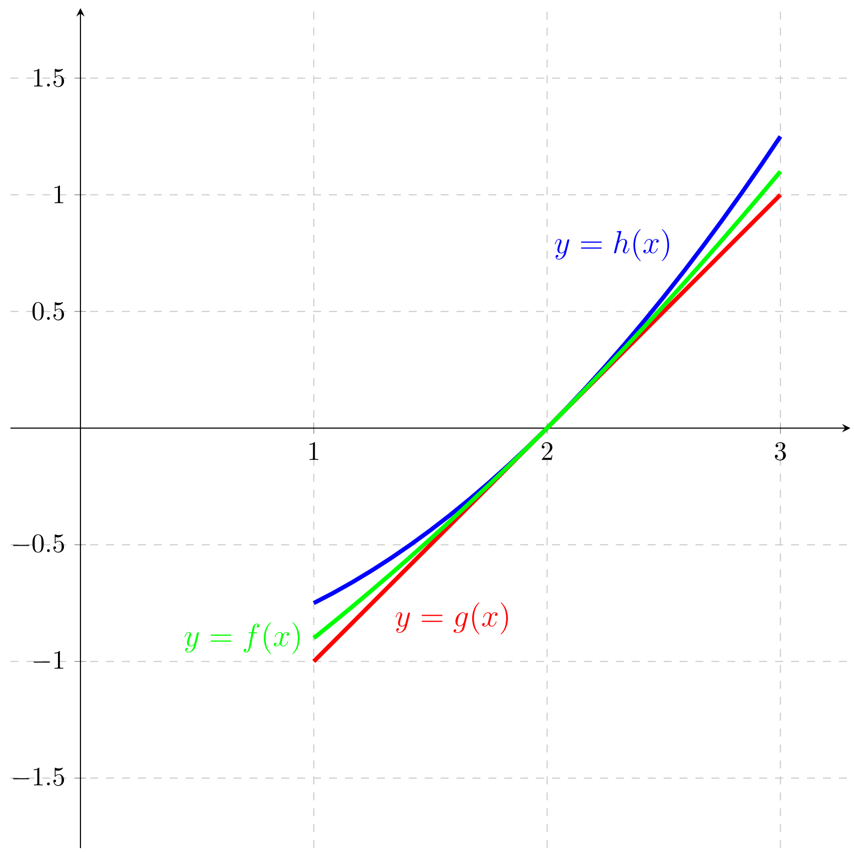
The graph of the function h is given in the figure below. Let f be a function defined on the interval $(1, 3)$ that satisfies the following inequalities

$$g(x) \le f(x) \le h(x),\ 1 < x < 3$$

(a) *In the figure below, sketch and label the graph of g and a possible graph of f. (All three functions have a common domain $(1, 3)$.)*



Explanation.



(b) *Evaluate the limit, or state that it does not exist. Justify your answer.*

(i) $\lim_{x \to 2} f(x)$

Explanation. g and h are both continuous function since they are both polynomials so we have:

$$\lim_{x \to 2} g(x) = g(2) = 0$$

$$\lim_{x \to 2} h(x) = h(2) = 0$$

We have $g(x) \leq f(x) \leq h(x)$ so by the squeeze theorem, $\lim_{x \to 2} f(x) = 0$ since $g(x) \leq f(x) \leq h(x)$ with $\lim_{x \to 2} g(x) = \lim_{x \to 2} h(x) = 0$.

(ii) $\lim_{x \to 2} \dfrac{f(x) + 2}{x - 1}$

Explanation.

$$\lim_{x \to 2} \frac{f(x) + 2}{x - 1} = \frac{\lim_{x \to 2} (f(x) + 2)}{\lim_{x \to 2} (x - 1)} = \frac{0 + 2}{2 - 1} = \frac{2}{1} = 2$$

(iii) $\lim\limits_{x\to 2} g(1+e^{f(x)})$

Explanation.

$$\lim_{x\to 2} g\Big(1{+}e^{f(x)}\Big) = g\Big(\lim_{x\to 2}(1{+}e^{f(x)})\Big) = g\Big(\lim_{x\to 2} 1{+}\lim_{x\to 2} e^{f(x)})\Big) = g\Big(1{+}e^{\lim\limits_{x\to 2} f(x)}\Big)$$

$$= g\Big(1+e^{0}\Big) = g(2) = 0$$